Version 1.0
Ref: 02002008/1.0/0

LOGO

Auto Dispatch Rules Specifications

Original Review and Approval				
	Name	Signature	Role	Date
Author (IMS):	Peter Kingston		Tech. Author / Bus. Analyst	20/04/10
Reviewer (IMS):	REMOVED		REMOVED	
Approver (IMS)	REMOVED		REMOVED	
Approver (COMPANY B)	REMOVED		REMOVED	
Approver (COMPANY B)	REMOVED		REMOVED	

The contents of this document are confidential and must not be communicated in whole or in part to any other party without the prior written approval of COMPANY A

Table of Contents

INTRODUCTION 5

1. OVERVIEW 6

2. SPECIAL RULES 7

3. STRATEGIES 10

3.1. Load Balance 10
3.2. Nearest Job 11
3.3. Fast Response 12
3.4. Zone Allocated 13

APPENDIX A: AUTO DISPATCH ENGINE - HIGH LEVEL FLOWCHART 14

APPENDIX B: INPUTS 15

APPENDIX C: REQUIREMENTS MATRIX 16

 © 2025 by COMPANY A

Revision History

Version	Date	Reason	Who
1.0 A Draft	20/04/10	Initial version	P Kingston

Distribution

	Name	Company
1	REMOVED	
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		

 © 2025 by COMPANY A

Confidentiality Notice

This document has been prepared by COMPANY A (“COMPANY A”) for the sole use of (“COMPANY B”). The contents are confidential and must not be communicated in whole or in part to any other party without the prior written approval of COMPANY A.

IMS has prepared this document in good faith based on the information made available to it. Many factors outside COMPANY A’s current knowledge or control may affect the recipient’s needs and plans. The statements in this document are qualified accordingly.

This document is provided subject to contract. Nothing in this document or in any related discussions or correspondence shall be construed as an offer, or the basis of any contract, or a representation which may be relied upon by any person except as COMPANY A may expressly agree in writing.

The following notice applies to this document and shall be reproduced on any permitted copies:

“Copyright © COMPANY A

All rights reserved.”

Introduction

Purpose of document

This document describes the various custom rules and default strategies to be used by the COMPANY A COMPANY B Solution Auto Dispatch Engine. These are placed in context of the end-to-end business process by the separate *COMPANY B Business Process* document and associated flowcharts. Within these flowcharts, an activity box colour coded as follows indicates that the Rule is described here.

Note: For an explanation of some of the terms used in this document, refer to the separate *COMPANY B Glossary*

Requirements

The functionality in this document addresses COMPANY B's requirements as specified in the separate document, *COMPANY B Requirements Specifications V1.21*. Where a piece of functionality described in this document directly satisfies a requirement from the Requirements Specifications, the associated requirement ID is included in square brackets thus **[R2]**. Refer to page 16 for a summary of requirements addressed in this document.

 © 2025 by COMPANY A

1. Overview

Auto Dispatch will be a key component of the COMPANY A solution. It will receive sets of data and then makes intelligent decisions based on various predefined rules, and groups of rules (*strategies*). Whilst the framework of the Auto Dispatch functionality (hereinafter, *the Auto Dispatch Engine*) is a fixed component of the solution, the associated rules and strategies can be tailored to individual business needs. For this reason, the actual rules for a specific customer are detailed here rather than in the separate *Auto Dispatch Functional Specifications* document.

The Auto Dispatch Engine can be used to determine the best individual or Team (hereinafter, *resource*) to deal with Work and will provide a single resource for information relating to the dispatching of Work. For each Work Order that is raised, the Engine will determine a list of resources who can carry out the work by applying basic rules. For example, these may include:

- Is the resource signed on for work?
- Does the resource have the skill set to deal with the issue? This might take into account not only resource capabilities, but also requirements for multiple resource attendance
- Does the resource’s current workload allow allocation of a new Work Order? This might take into account the amount of work the resource has accepted recently, and the length of time the resource has remaining on shift.
- Is the resource's geographical location (determined by GPS) within reasonable travelling time of the work Site? This will take into account site opening and closing times, the associated Service Level Agreement

Further, more granular rules can be then be applied to the resultant list of resources to determine the most appropriate resource for the job. The Dispatch Engine will then match an appropriate resource to the Work by first filtering out resources that do not qualify - for example, those that are not on shift, or do not have the relevant skills, or have the relevant skills but do not have enough shift time left to reach the site and complete the Work.

Resources that make it through the first filter are then weighted against the chosen strategy (or group of rules) for the Branch, region or department with which the site is associated, and a ‘best fit’ resource is selected. Should there be several resources that match the required criteria, then these will be sorted by their respective ETA so that the nearest resource is assigned the task. The COMPANY A solution will then automatically dispatch the allocated Work to the selected resource via a PDA. However, Agents may have to intervene in the process where Work Orders have a status requiring manual intervention.

 © 2025 by COMPANY A

2. Special Rules

To support Auto Dispatch, there will be customisable dispatch rules and methodologies (for example driven by best engineer / location and zoning) **[R16]**

In addition to the default strategies described in the next section, the Auto Dispatch Engine must take account the following special rules identified as requirements by COMPANY B in the document *COMPANY B Requirements Specifications V1.1*.:

1. Work Orders created for the provision of CIT deliveries will be passed to the COMPANY A system and for the purposes of automated dispatch will have a set schedule and route associated to them. CIT team routes will be passed to the dispatch engine based on the data from MSD. It will be possible to view this route on the COMPANY A automated dispatch interface **[R53]**, but they will not be auto dispatched as CIT Teams do not currently have PDAs. Such Work Orders will be placed in a status that requires the Dispatch Agent to investigate and then manually assign the work.
2. If an ATME call is placed which appears to be on a CIT route, an assumption of time per location will be used to attempt manual dispatch of the relevant ATME call to the CIT team if they are at the assumed location. The route will either contain ETAs for each ATM or a sequence in which the ATMs will be visited. Each route will have a start and end time **[R87]**.
3. Meets and project works will be treated as Work Orders and retrieved from MSD **[R90]**
4. As keys are carried by the CIT team for the route they are attending to on any given day, field engineering teams must be provided with the relative key holding data as part of the business process **[R54]**.
5. Currently, COMPANY B makes use of Teams of two engineers to attend to faults. Due to the engineers working in Teams, the PDA will be allocated to the Team and Auto Dispatch must treat the Team as a single resource **[R55]**. For clarity, each PDA will be configured with the ID of a specific Route. The Team allocated to this route will be given the appropriate PDA for that Route at the start of their shift and Auto Dispatch will treat them as a single resource.
6. However, in some instances, Teams may be split if non-vault work can be undertaken and a remaining Team member will continue with one-person jobs until the other team member has finished and can return to route **[R56]**.
7. A Team is allowed to enter a different region other than their own to attend a Work Order, either because they have spare keys; or do not require keys for the associated ATM. However a Team from a neighbouring region cannot go to the Branch to collect keys for that region **[R58]**

 © 2025 by COMPANY A

8. Project Teams will be assigned maintenance Work Orders only if their scheduled work is completed with sufficient time left on shift to attend the Work Order **[R59]**
9. Scheduled/Project teams will not be re-routed as scheduled work takes priority **[R60]**
10. Bank holiday routes are deployed only if requested by customer **[R61]**
11. Ink dye ATMs can be attended to by one engineer whether it is a vault or non-vault issue **[R74]**
12. There are certain known sites that have access issues and so require the use of emergency keys. A call will be logged once and if a subsequent call is raised due to access issues the team will not be sent again without an emergency key **[R79]**. If a Work Order is for an ATM that requires emergency keys, then the dispatch will be managed manually and a decision made depending on the estimated location of the CIT team and whether an ATME team can collect the keys from CIT and attend within SLA **[R77]**. The requirement for an emergency key will be flagged against the ATM in MSD and passed to COMPANY A in the overnight upload.
13. Auto Dispatch must support performance of project work to a pre-agreed schedule for their customers and these will usually consist of a third party meet. MSD will always push Project Routes out only for the following day regardless of when it was created in MSD. This will include the Route ID and the start and end times and sequence number **[R80]**.
14. Route skill sets do not change and so appropriately skilled engineers will be assigned each day **[R89]**
15. If a CIT Route is assigned an ATME Work Order, Auto Dispatch will realign the ETA times for the remaining stops on the CIT Route **[R94]**
16. If FLM Work Order takes longer than 20 minutes the team must contact customer's helpdesk and advise. The Team must also record who they have spoken to **[R96]**.
17. Lobby ATMs and ink dye machines can be attended by a single engineer ('split Team'). Any fault that is a 'top' fault can also be attended by a split Team. Access restrictions apply to both lobby and post office ATMs. Single engineers will be identified by a skill type of "single" **[R99]**.
18. If a Team is due to meet a third party and the third party does not attend within 15 minutes of agreed time, the team must contact the Dispatch Team who will contact third party. If the time needs to be rearranged, the current call is closed and a new one opened. The Work Order is to be closed by Dispatch Team **[R101]**.
19. Auto Dispatch will need to take into account that Teams are connected to one of five regions. One Team will have the keys to enable them to work in a neighbouring region if required. This is the same route and skill set everyday **[R107]**.
20. Any Team will be able to go into any region as long as the access method is keyless. A slide bar will be added to Auto Dispatch to enable the

 © 2025 by COMPANY A

sensitivity of distance travelled to be controlled – default will be 50 km. **[R108]**

21. Teams can also be assigned to work within another region for the day. This will be achieved by altering the Team Branch for the day **[R109]**.
22. If Work Order is for a Meet or a Project that requires a 3rd Party, then it will be assumed that the Work is FLM and it will be dispatched accordingly
23. If a Work Order is for an ATM that has special access instructions then the Work Order will be flagged with “Manual Dispatch” requirements
24. Engineers are able to reject a work order if they are being sent too far away from home area (this is not a regional boundary strategy but a total distance travelled strategy)
25. For the purposes of Auto Dispatch, all Teams contain two people and always start work from the home branch (even when covering different branches)

 © 2025 by COMPANY A

3. Strategies

By default the COMPANY A solution will offer the following strategies per customer Branch. For details of how each strategy can be applied, refer to the separate *COMPANY B Auto Dispatch Functional Specifications*:

3.1. Load Balance

3.1.1. Aim

Ensure there is an even distribution of Work Orders among the Teams that are on shift. Once qualifying Teams have been identified, then the Team with the least Work Orders currently assigned will be assigned the next Work Order.

3.1.2. Example

Work Order 555666 is to be dispatched. Three Teams have qualified:

- Team A has two Work Orders assigned
- Team B has three Work Orders assigned
- Team C has one Work Order assigned

Result: Work Order 555666 is dispatched to Team C

 © 2025 by COMPANY A

3.2. Nearest Job

3.2.1. Aim

Ensure that new Work Orders are assigned to the closest Team with a current Work Order. The GIS position of the ATM on the Work Order in question is compared against the GIS positions of the ATMs on all the other assigned jobs for the branch. The Team with the closest Work Order to the Work Order in question will be assigned to it.

3.2.2. Examples

Work Order 123456 is to be dispatched and has a GIS position of X. Three Teams have qualified, each already having one Work Order assigned to them.

- Team A's assigned Work Order is 20 km from Work Order 123456
- Team B's assigned Work Order is 5 km from Work Order 123456
- Team C's assigned Work Order is 7 km from Work Order 123456

Result: Work Order 123456 is assigned to Team B as their currently assigned Work Order is the closest.

In the event that more that one Team fulfils all the strategy criteria for the Nearest Job strategy, a *Default* strategy is applied to determine which Team is to be given the Work Order. The strategy works in the following order:

1. If any Team is within the Nearest Override setting for the Branch (e.g. within 500 m) they will be selected
2. If there are multiple Teams within the Nearest Override then the Teams are weighted on which has the least number of Work Orders
3. If there are multiple Teams with matching least number of Work Orders then the Teams are weighted on the Nearest Job strategy as described above
4. If there are still multiple qualifying Teams then the Fast Response strategy is used (see page 12)

 © 2025 by COMPANY A

3.3. Fast Response

3.3.1. Aim

To dispatch a Team to site as quickly as possible (i.e., from among the qualifying Teams, assign to the Team that will arrive on site first). This takes into account the current work load of each qualifying Team and the distance between each Work Order in their current work load.

Note: Note: If Teams last GPS positional reading is older than 60 minutes then the configuration setting of the average travel time is used instead.

3.3.2. Example

Work Order 654321 is being dispatched. Two Teams qualify:

- Team A has two Work Orders assigned
- Team B has three Work Orders assigned

The configuration settings for the Branch are (see the separate *COMPANY B Auto Dispatch Functional Specifications*):

- Average Meet job is set to 40 minutes
- Average FLM job is set to 15 minutes

Team A

Work Order	Average Job Time	Travel Time to next	WO Total
100001 (Meet)	40 minutes	25 minutes	65 minutes
100002 (Meet)	40 minutes	20 minutes	60 minutes
654321			125 minutes

Team B

Work Order	Average Job Time	Travel Time to next	WO Total
100003 (FLM)	15 minutes	10 minutes	25 minutes
100004 (Meet)	40 minutes	5 minutes	45 minutes
100005 (FLM)	15 minutes	15 minutes	30 minutes
654321			100 minutes

Result: Work Order 654321 is dispatched to Team B as their combined timings for current workload is less than Team A – and they would therefore arrive at the Work Order first

 © 2025 by COMPANY A

3.4. Zone Allocated

3.4.1. Aim

Dispatch Work Orders to Teams that already have Work Orders assigned within the same zone (for more on zoning see the separate *COMPANY B Auto Dispatch Functional Specifications*):

3.4.2. Example

Work Order 232323 is to be dispatched. The ATM has an associated zoning of 1. Three Teams qualify. They have the following Work Orders already assigned:

- Team A's 1st assigned Work Order has no zoning applied to it
- Team A's 2nd assigned Work Order has a zoning of 1
- Team B's 1st assigned Work Order has no zoning applied to it
- Team B's 2nd assigned Work Order has a zoning of 2
- Team B's 3rd assigned Work Order has a zoning of 2
- Team C's 1st assigned Work Order has a zoning of 4
- Team C's 2nd assigned Work Order has a zoning of 4
- Team C's 3rd assigned Work Order has a zoning of 4

Result: Work Order 232323 is dispatched to Team A as one of the Team's already assigned Work Orders has a zoning of 1 which matches

In the event that more that one Team fulfils all the strategy criteria for the Zone Allocated Job strategy, a Default strategy is applied to determine which Team is to be given the Work Order. The strategy works in the following order:

1. If any Team is within the Nearest Override setting for the Branch (e.g. within 500 m) they will be selected
2. If there are multiple Teams within the Nearest Override then the Teams are weighted on which has the least number of Work Orders
3. If there are multiple Teams with matching least number of Work Orders then the Teams are weighted on the Nearest Job strategy (see page xx)
4. If there are still multiple qualifying Teams then the Fast Response strategy is used (see page 12)

 © 2025 by COMPANY A

Appendix A: Auto Dispatch Engine - High Level Flowchart

The following is a summary flowchart showing the sequence of decisions the will be made by the Auto Dispatch functionality when allocating a Work Order to a Team:

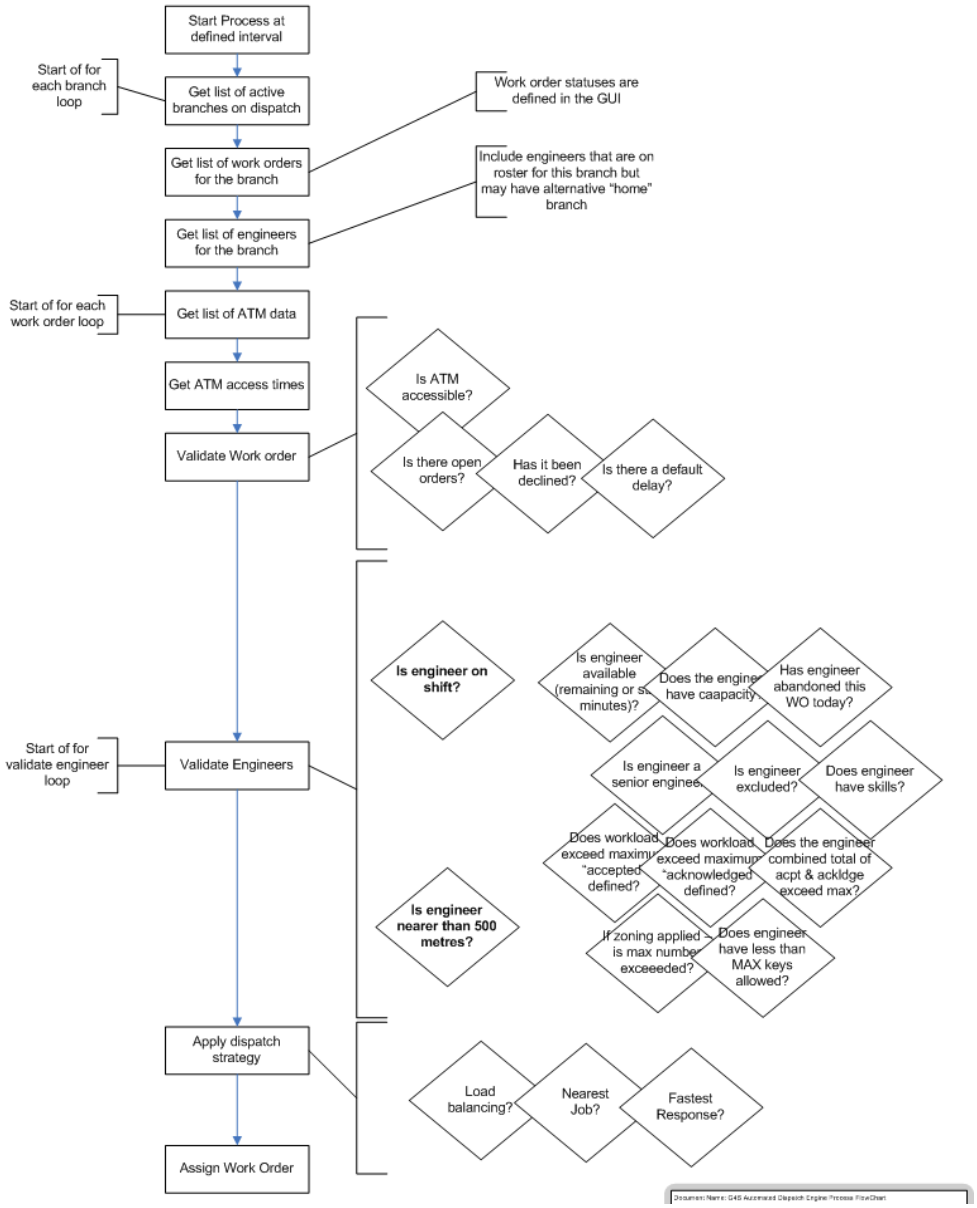

Figure 1: Auto Dispatch Engine - High Level Flowchart

Appendix B: Inputs

For clarity, the decisions using the Auto Dispatch Rules will be made with the use of the following data inputs:

Input	Source
Estimated geographic position of CIT Teams	Routes and ETAs uploaded from MSD
Actual geographic position of ATME Teams (Routes)	GPS data from PDA
Actual geographic position of Teams (Routes) performing Project Work	GPS data from PDA
Actual geographic position of Teams (Routes) assigned to Meets	GPS data from PDA
Type of Work Order and the requirements (one engineer / two engineer)	MSD
Which Team have which ATM keys	MSD
Details of the SLA associated with the Work Order	MSD
Which ATM sites have special conditions around access methods (site access)	MSD
Whether a Team allocated Project work has completed their schedule and still has shift hours left	COMPANY A
ATM access methodology – keyless etc.	MSD
Which sites require Emergency Keys	MSD
Distance an Engineer can travel from current location	COMPANY A
Average or calculated job times plus travel times	MSD

 © 2025 by COMPANY A

Appendix C: Requirements Matrix

The following is a summary of the requirements from *COMPANY B Requirements Specifications V1.1* addressed in this document.

RULES

R		R		R		R		R	
1		11		21		31		41	
2		12		22		32		42	
3		13		23		33		43	
4		14		24		34		44	
5		15		25		35		45	
6		16	✓	26		36		46	
7		17		27		37		47	
8		18		28		38		48	
9		19		29		39		49	
10		20		30		40		50	

R		R		R		R		R	
51		61	✓	71		81		91	
52		62		72		82		92	
53	✓	63		73		83		93	
54	✓	64		74	✓	84		94	✓
55	✓	65		75		85		95	✓
56	✓	66		76		86		96	✓
57		67		77	✓	87	✓	97	
58	✓	68		78		88		98	
59	✓	69		79	✓	89	✓	99	✓
60	✓	70		80	✓	90	✓	100	

R		R		R		R		R	
101	✓								
102									
103									
104									
105									
106									
107	✓								
108	✓								
109	✓								

 © 2025 by COMPANY A

Send Us Your Comments

Auto Dispatch Rules Specifications Version 1.0

COMPANY A welcomes your comments and suggestions on the quality and usefulness of this document. Your input is an important part of the information used for revision.

- Did you find any errors?
- Is the information clearly presented?
- Do you need more information? If so, where?
- Are the examples correct? Do you need more examples?
- What features did you like most?

If you find any errors or have any other suggestions for improvement, please indicate the document title and version number, and the part, section, and page number (if available). You can send comments to us in the following ways:

- Email: **peter.kingston@COMPANYA.com**
- Telephone +44 (0)1 and ask to speak to the Documentation Manager
- Post to:

COMPANY A.

 © 2025 by COMPANY A

REMOVED

Copyright © 2025 by COMPANY A. All rights reserved. All trade names are registered trademarks of the respective manufacturers.